

DECONFOUNDING USING THE PROBABILITY INTEGRAL TRANSFORMATION

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Summary: In this paper we consider the situation of before and after measurements on a treatment and a control case, where the effect of the treatment is confounded with some other factor. We show how the probability integral transformation can be used to carry out the deconfounding when one has a very large sample. We apply the method to data from a radar experiment that originally motivated the study.

1. INTRODUCTION

Consider the well-known situation where we want to study the effect of a treatment or factor. We do this by sampling data before and after the treatment has been applied. Unfortunately, in many experiments one cannot isolate the effect of the treatment since the latter is confounded with another factor. For example, there might be different environmental conditions prevailing at the time of the before and after measurements. Note that the confounding factor (environmental conditions) cannot be controlled by the experimenter.

One way of overcoming confounding is to have a control experiment where data are also sampled "before" and "after", but without the treatment having been administered. Then the change observed between the before and after data can be considered as the effect of the confounding factor. This information on the effect of the confounding factor can then be

used to deconfound the treatment case. A typical way of doing this is to model the effect of the confounding factor as a transformation defined by a small number of parameters, often a location or scale parameter. This may not be sufficient for many situations, but often the small sizes of the data sets do not allow one to do more.

In recent times recording equipment (usually electronic) has improved dramatically, often recording very large numbers of data automatically. In the above situation this means that one can model the effect of a confounding factor in a much more complex way, i.e. by a transformation defined by a large number of parameters. In the present paper we will use the probability integral transformation to model this transformation, thus effectively defining it by infinitely many parameters. We apply the transformation to data from a radar experiment which originally motivated the study.

In the next section this radar experiment is discussed, and in section 3 a solution is described and applied to the example. Some further work is discussed in section 4, where the case of smaller samples is considered.

2. PROBLEM DESCRIPTION

Radar is a ship's main sensor for detecting surface or airborne objects. An active radar transmits electromagnetic signals, which echo off the target. The echoed signal is received and interpreted, giving for example the radar size of the object.

In a military situation one wants a small radar size for one's own ship, since then it is more difficult to be "seen". There are different ways of modifying a ship to reduce its radar size. One possibility is to stick radar-absorbing material onto the ship. This material partially absorbs the incoming signal and reduces the echo. Another possibility is to scatter the echo signal, thus creating a confused radar picture.

When testing the effectivity of a modification, the question to be answered is whether the modification reduced the radar size. There are many different criteria that can be used to try to answer this question. In

our case the quantity of interest to the engineers working on the problem was the probability that the echo signal after modification was less than the echo signal before modification. We had thus to design and carry out an experiment to estimate this probability. Unfortunately a number of weeks elapsed between the before and after modification measurements which meant that the environmental conditions were quite different for the two sets of measurements. The environmental conditions thus acted as a confounding factor which had to be removed before a valid comparison could be made. To overcome the confounding, a control ship was used on which measurements were made simultaneous with the before and after measurements on the experimental ship. The control ship was of course not modified.

In the next section we describe our deconfounding method and apply it to the above example.

3. SOLUTION AND APPLICATION

Let X_B and X_A be random variables representing respectively the before and after measurements on the treatment, and similarly Y_B and Y_A for the control. Let also X_B^* be an "adjusted" before measurement, adjusted through the "adding" of the environmental effect to X_B (how this is done is described below). We are then interested in

$$p \equiv P(X_A < X_B^*).$$

A value of p "close" to one means that the treatment was successful. A value of p close to $\frac{1}{2}$ means the treatment had no effect.

Denote the distribution functions of X_B , X_A , Y_B , Y_A and X_B^* respectively by G_B , G_A , F_B , F_A and G_B^* and the corresponding densities by lower case letters.

By the probability integral transformation (see Randles and Wolfe, 1979) we can write

$$Y_A \stackrel{D}{=} F_A^{-1}(F_B(Y_B)) \approx \gamma^{-1}(Y_B)$$

where $\stackrel{D}{=}$ means equal in distribution. We can say that the change in environment is modelled by γ^{-1} . To "add" the environment to X_B we apply γ^{-1} to X_B to obtain X_B^* , i.e.

$$X_B^* = \gamma^{-1}(X_B).$$

Now,

$$\begin{aligned} G_B^*(x) &\equiv P(X_B^* \leq x) \\ &= P(\gamma^{-1}(X_B) \leq x) \\ &= P(X_B \leq \gamma(x)) \\ &= G_B(\gamma(x)), \end{aligned}$$

and in terms of densities

$$\begin{aligned} g_B^*(x) &= dG_B^*(x)/dx \\ &= g_B(\gamma(x)) \cdot \gamma'(x) \\ &= g_B(F_B^{-1}(F_A(x))) \cdot f_A(x) / f_B(F_B^{-1}(F_A(x))) \end{aligned}$$

In terms of the quantity of interest, we have

$$\begin{aligned} p &= P(X_A < X_B^*) = \int P(X_A < x) dG_B^*(x) \\ &= \int G_A(x) dG_B^*(x) \\ &= \int G_A(x) g_B^*(x) dx \end{aligned}$$

with g_B^* given by the above expression.

If we know the different distributions involved in the above expression, we can calculate p exactly. If not, sample observations have to be used to estimate p .

We now return to the radar example of the previous section. In that experiment we had very large samples so that the above distributions were essentially known, and p could be calculated exactly. We now describe the experiment in more detail.

Two similar ships were used, one as the target (treatment) and one as the control. Radar signals were transmitted to one ship at a time, at a fixed distance and ship angle. The echo signal from the ship was recorded. For each ship this was done for four minutes at a rate of 64 hz, thus giving two sets of observations of size $4 \times 60 \times 64 = 15\,360$ each. This was repeated for 19 distance and angle combinations. The measurements on the two ships were made at approximately the same time so that they were made under essentially the same environmental conditions.

The target ship was then modified and after three weeks the measurement process on the two ships was repeated. This again yielded two sets of observations of 15 360 each. Thus for each distance and angle combination we have the following data:

	TARGET	CONTROL
Before	15 360 obs. on X_B	15 360 obs. on Y_B
After (3 weeks)	15 360 obs. on X_A	15 360 obs. on Y_A

The effect of the modification is confounded with the effect of the environmental change. The standard practice is to simply look at the change in means, i.e. at a location change, and to deconfound on that. Because we had such large samples, we felt that one could do better than simply looking

at a location change, and we proposed the above solution.

Kernel density estimation subroutines of the IMSL library (9th ed., 1982) were used to calculate f_B , f_A , g_B , g_A , g_B^* . The calculated densities were only positive over a finite interval and g_B^* was calculated as follows: Let f_B be defined on (a,b) , f_A on (c,d) and g_B on (e,f) , and define the location-scale transformation:

$$\delta^{-1}(x) = (x - \alpha)/\beta$$

with α and β chosen so that $\delta^{-1}(\alpha) = e$ and $\delta^{-1}(b) = f$

Then

$$h_B(x) = g_B(\delta^{-1}(x))$$

is a scaled version of g_B and defined on the interval (a,b) . The transformation γ is defined on (c,d) with $\gamma(c) = a$ and $\gamma(d) = b$ and when h_B is transformed by γ we have

$$h_B^*(x) = h_B(\gamma(x)) \cdot \gamma'(x)$$

with $h_B^*(x)$ defined on (c,d) . The inverse location-scale transformation $\delta(x) = \alpha + \beta \cdot x$ is then applied to obtain

$$\begin{aligned} g_B^*(x) &= h_B^*(\delta(x)) \\ &= h_B(\gamma(\delta(x))) \cdot \gamma'(\delta(x)) \\ &= g_B(\delta^{-1}(\gamma(\delta(x)))) \cdot \gamma'(\delta(x)) \end{aligned}$$

with g_B^* defined on $(\delta^{-1}(c); \delta^{-1}(d))$. Using these results, the value of p was calculated using numerical integration.

EXAMPLE: Consider the distance/angle combination (15 km, 20°). Figure 1 shows the densities f_B , f_A , g_B and g_B^* .

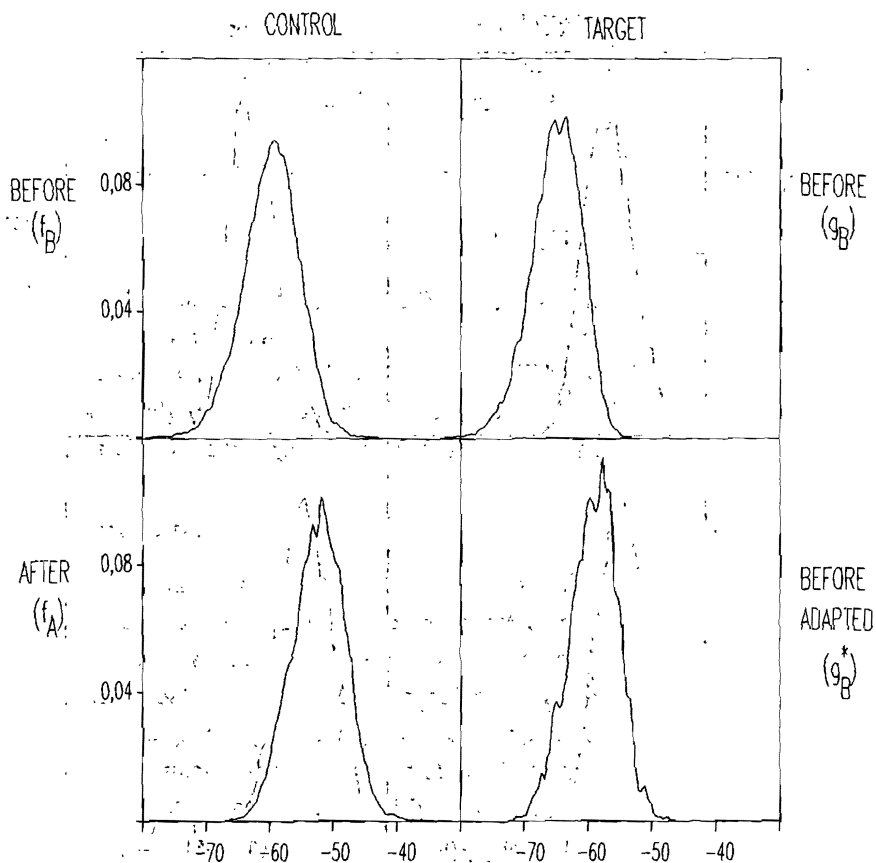


Figure 1. Densities f_B , f_A , g_B and g_B^* for the distance/angle combination (15 km, 20°).

Figure 2 gives the density g_A compared with g_B and g_B^* .

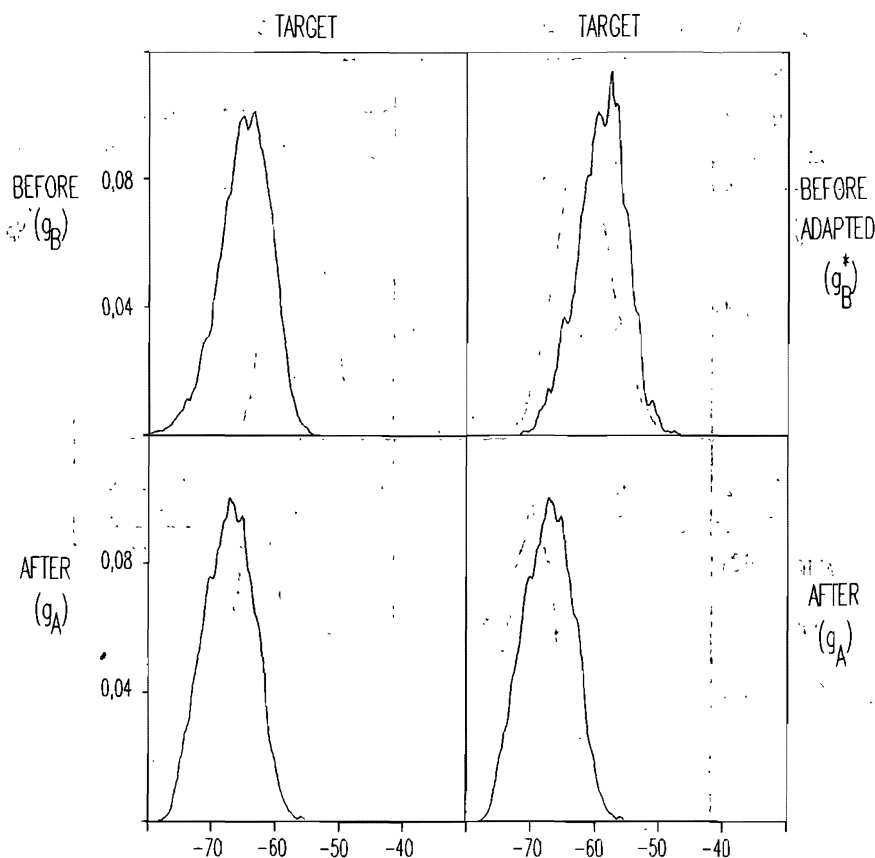


Figure 2. Comparison of g_A with g_B and g_B^* for the combination (15 km, 20°).

We see that in this example the echoes from the control ship after the treatment, Y_A , are on average larger than Y_B . The echoes X_B are transformed and X_B^* values are on average larger than the X_B 's. In figure 2 we

see that when comparing g_A with g_B the effect of the treatment (confounded with the effect of the environment) is not that obvious. When the density g_B^* is compared with g_A , i.e. the effect of treatment compared with no treatment, we see that the treatment reduced the radar size on average.

Calculations for this example give

$$p = P(X_A < X_B^*) = 0,93.$$

This value obtained for p means that the modification was successful. We note that if we compare the "before" and "after" directly, i.e. without deconfounding, we find that this gives $P(X_A < X_B) = 0,66$. This is substantially different to the value of 0,93 obtained by first deconfounding. Other distance and angle combinations gave similar results.

4. SOME REMARKS

- 4.1 Sometimes the sample sizes are not large enough to calculate p exactly. In such a situation p has to be estimated and thus sampling variability must be taken into account. We then obtain $\hat{p}$ as an estimate of p and would also want a confidence interval on $\hat{p}$. When the sample sizes are large enough for asymptotic theory to apply we can obtain an approximate confidence interval as follows. Denoting all estimated distributions as before but with a " $\hat{\cdot}$ " on, we have

$$\hat{p} = \int \hat{G}_A(x) d\hat{G}_B^*(x)$$

and thus

$$\hat{p} - p = \int (\hat{G}_A(x) - G_A(x)) d\hat{G}_B^*(x) + \int \hat{G}_A(x) d(\hat{G}_B^*(x) - G_B^*(x))$$

$$= \int (\hat{G}_A(x) - G_A(x)) d\hat{G}_B^*(x) + \int G_A(x) d(\hat{G}_B^*(x) - G_B^*(x)) + R$$

where

$$R = \int (\hat{G}_A(x) - G_A(x)) d(\hat{G}_B^*(x) - G_B^*(x)).$$

Under reasonable assumptions we have

$$R = O_p(n^{-1}), \text{ as } n \rightarrow \infty,$$

where n is the sample size of each set of observations (see, e.g. Shorack and Wellner, 1986, for results of this type). For the other two terms we have, again under reasonable conditions, by the central limit theorem that

$$\sqrt{n} \int (\hat{G}_A(x) - G_A(x)) dG_R^*(x) \xrightarrow{D} N(0, \sigma_1^2)$$

and

$$\sqrt{n} \int G_A(x) d(\hat{G}_B^*(x) - G_B^*(x)) \xrightarrow{D} N(0, \sigma_2^2)$$

where $\xrightarrow{D}$ denotes convergence in distribution. Here

$$\sigma_1^2 = \int \int (G_A(x) \wedge G_A(y) - G_A(x)G_A(y)) dG_B^*(x) dG_B^*(y)$$

and

$$\sigma_2^2 = \int \int (G_B^*(x) \wedge G_B^*(y) - G_B^*(x)G_B^*(y)) dG_A(x) dG_A(y)$$

with $a \wedge b = \min(a, b)$.

Since $\hat{G}_A$ and $\hat{G}_B^*$ are uncorrelated, it follows from the above that

$$\sqrt{n}(\hat{p} - p) \xrightarrow{D} N(0, \sigma^2)$$

where

$$\sigma^2 = \sigma_1^2 + \sigma_2^2.$$

Estimating the distribution functions in the expressions for σ_1^2 and σ_2^2 by the corresponding empirical distribution functions, we get an estimate for σ^2 , say $\hat{\sigma}^2$.

As a confidence interval we can then use

$$\hat{p} \pm z\hat{\sigma}/\sqrt{n},$$

where z is the appropriate normal quantile. See also Govindarajulu (1968).

Note that one will only apply the present technique in cases where one has large samples. In cases of small samples one can still in principle apply the technique and use a resampling method (e.g. the bootstrap, see Efron, 1979) to find a confidence interval for p .

- 4.2 Another example, in the medical field, where this technique was applied is the analysis of data from a study of coronary risk factors in the populations of three towns. Over a period of four years the populations of two of the towns were subjected to intervention strategies aimed at reducing the risk of coronary heart disease, by means of lowering blood pressure, adopting a prudent diet, etc. The data from the third town (the control), representing the effects of environment and ageing, were used to modify the data of the intervention towns by means of the techniques of this paper, and to provide a measure of the effect of intervention (see Chalton, 1989).
- 4.3 In the radar example it is to be expected that there will be some dependence between the data points. Autocorrelation plots indicate the presence of some autocorrelation within the different sequences. We are still interested in $P(X_A < X_B^*)$, where this is now defined in terms of the marginal distributions of the relevant (dependent) sequences. Implicitly we thus assume that the marginal distributions of each sequence are identical, which is a reasonable assumption to make in our case. Also, the data should be sufficient to give (estimate) the relevant distribution functions, which was again in order in our case.

REFERENCES

- CHALTON, D.O. (1989). Private communication.
- EFRON, B. (1979). Bootstrap methods: another look at the jackknife. *Ann. Statist.*, 7, 1-26.
- GOVINDARAJULU, Z. (1968). Distribution-free confidence bounds for $P(X < Y)$. *Ann. Inst. Stat. Math.*, 20, 229-238.
- RANDLES, R.H. & WOLFE, D.A. (1979). *Introduction to the theory of nonparametric statistics*. John Wiley & Sons, New York.
- SHORACK, G.R. & WELLNER, J.A. (1986). *Empirical processes with applications to statistics*. John Wiley & Sons, New York.